

Ashank Dsouza

(480) 277-5451 · ashankdesouza@gmail.com · linkedin.com/in/ashank-d/ · Tempe, AZ, USA

PROFESSIONAL SUMMARY

Forward-deployed engineer with 5+ years shipping production systems across ML infrastructure, high-traffic consumer platforms, and distributed microservices. Founding engineer at PangeaTech — owned end-to-end client deployments, cut data processing by 80%, and reduced deployment failures to near-zero via Docker + CI/CD automation.

TECHNICAL SKILLS

AI & LLM Integration: Claude AI (Anthropic), GitHub Copilot (AI-agent workflows), LLM API integration, Prompt Engineering

Customer Deployment & Integration: Python, FastAPI, Node.js, Express, TypeScript, REST APIs, Microservices, System Integration

Frontend & Dashboards: React, Redux, D3.js, JavaScript, HTML, CSS

Data: PostgreSQL, MySQL, MongoDB, SQL, Elasticsearch, Tableau (Materialized Views)

Cloud & Infrastructure: AWS (EC2), Azure (VM), Docker, Kubernetes, GitHub Actions, Git, Linux

Observability & Reliability: EFK Stack (Elasticsearch, Fluentd, Kibana), Centralized Logging, 99%+ SLA maintenance

EXPERIENCE

Senior Software Engineer · 04/2023 – 12/2023 · **Tecnotree** · Bengaluru, KA

- **Awarded the "Spotlight" cash award** for leading high-quality, on-time feature delivery in React and Node.js — recognized for production impact and engineering excellence.
- Owned end-to-end development of customer-facing UI screens using reusable React components and Node.js APIs, directly improving client product quality in a live production environment.

Senior Software Engineer · 07/2022 – 03/2023 · **PangeaTech** · Bengaluru, KA

Founding engineer — owned backend performance, infrastructure, and client-facing deployment of an ML product.

- **Delivered 80% reduction in client data processing time** (4 hrs → 45 min) by redesigning the backend to distribute ML workloads across Kubernetes-orchestrated containers — resolving a critical client bottleneck.
- **Eliminated 4-hour lag in bug diagnosis** across distributed client systems by deploying centralized EFK stack observability, enabling real-time root cause identification and dramatically improving SLA response.
- **Cut client Azure cloud spend by 30%** through strategic VM utilization optimization via multi-core Node.js scaling — translated to direct cost savings presented to client stakeholders.
- Mentored junior engineers through 50+ code reviews, establishing coding standards that improved team delivery velocity and codebase maintainability.

Software Engineer · 11/2021 – 07/2022 · **PangeaTech** · Bengaluru, KA

- **Engineered a 97% reduction in client MySQL refresh cycles** (8 hrs → 15 min) by rewriting synchronous Python to async — directly eliminating hours of weekly client downtime.
- Maintained 50+ SQL materialized views powering client Tableau business-intelligence dashboards, ensuring data accuracy and availability for stakeholder reporting.
- Ensured 99% MySQL database availability via server redundancy, upholding client SLA commitments for mission-critical data systems.
- **Reduced deployment time by 85%** (27 → 4 min) and near-eliminated deployment failures by containerizing the application with Docker and automating CI/CD via GitHub Actions.

Software Engineer · 07/2019 – 04/2021 · **Tenerity (a JPMorgan Chase Company)** · Pune, MH

- Built and delivered production REST APIs and customer-facing frontend components for a **high-traffic flight booking platform serving millions of customers** — end-to-end ownership from design to deployment.
- Raised test coverage to 90% and enforced WCAG accessibility and internationalization standards, ensuring enterprise-grade reliability and compliance for a regulated financial product.
- Applied TDD, OOP, and SOLID principles to deliver maintainable, production-grade code across a large-scale distributed system.

ADDITIONAL EXPERIENCE

Team Lead — Engineering · 08/2025 – 12/2025 · **ABC Academy of Music (Email Marketing Microservice)** · Remote

- Led a 5-person engineering team through full delivery cycle — requirement gathering with the product owner, technical decision-making, and sprint execution.
- **Boosted team development velocity by 75%** by introducing AI-agent-based workflows using GitHub Copilot, demonstrating hands-on LLM tooling adoption in a production team.

- Built Java Spring Boot REST APIs backed by PostgreSQL; automated CI/CD pipeline via GitHub Actions on AWS EC2, achieving 99% uptime.

Fullstack Engineer · 09/2021 – 10/2021 · **Friendly Brands (Upsell Widget UI)** · Remote

- **Improved e-commerce click-through rate by 30%** via data-driven A/B testing of alternative UI designs — translated customer behavior insights directly into shipped product changes (FastAPI, Angular, SSR).

Engineering Tutor · 08/2024 – 12/2025 · **Arizona State University** · Tempe, AZ

- Delivered 100+ tutoring sessions on Agile, Data Structures & Algorithms, Operating Systems, and Databases — strengthening communication of complex technical concepts to diverse audiences.

PROJECTS

Web Simulator · Built with Claude AI (Anthropic)

- Designed and built a programmatic system simulation tool capable of modelling any target system, featuring real-time interactive visualisations via D3.js and echarts.
- **Developed end-to-end using Claude AI (Anthropic)** — demonstrating hands-on LLM-assisted rapid prototyping, from architecture design to iterative feature implementation with AI collaboration.
- Showcases practical AI integration capability: leveraging LLMs as active engineering collaborators to ship working software faster.

EDUCATION

Master of Science in Software Engineering · 3.87 / 4.0 GPA · 01/2024 – 12/2025 · Arizona State University, Mesa, AZ

Coursework: System Design (Distributed & Scalable Microservices), Data Structures & Algorithms, UX

Bachelor of Science in Computer Science · 08/2019 · REVA University, Bengaluru, KA